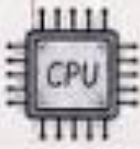


“ Understand the structure. ”
 ★ Design the Future. ★

COMPUTER ARCHITECTURE

★ NOTES ★

— Study • Understand • Build —




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• CPU
• Memory
• I/O
• System Bus

WHAT IS COMPUTER ARCHITECTURE?

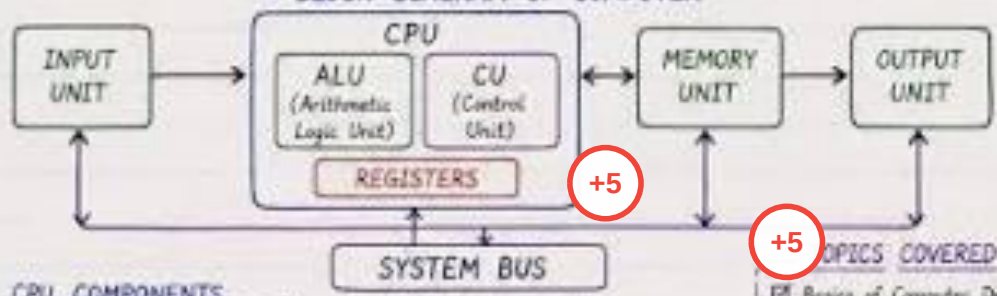
Computer Architecture refers to the design and structure of a computer system. It deals with the functional units, their interconnections and their working together to execute instructions.



MAIN COMPONENTS

- CPU (ALU, CU, Registers)
- Memory System
- I/O Modules
- System Interconnection

BLOCK DIAGRAM OF COMPUTER



CPU COMPONENTS

- ★ **ALU** – Performs arithmetic and logic operations.
- ★ **CU** – Controls all operations and coordinates all parts.
- ★ **Registers** – Small, fast storage locations inside the CPU.

TOPICS COVERED

- ☒ Basics of Computer Design
- ☒ Number Systems
- ☒ Boolean Algebra
- ☒ Logic Gates
- ☒ Combinational Circuits
- ☒ Sequential Circuits
- ☒ CPU Organization
- ☒ Instruction Formats
- ☒ Addressing Modes
- ☒ Memory Organization
- ☒ I/O Organization
- ☒ System Bus

MEMORY HIERARCHY



Faster
Smaller
Costlier
↓
Slower
Larger
Cheaper

“ Architecture is the foundation,
Performance is the result. ”

@mastercode.sagar

Good Architecture
Makes Powerful Systems.

★

RAM (RANDOM ACCESS MEMORY)

Memory Architecture & Working

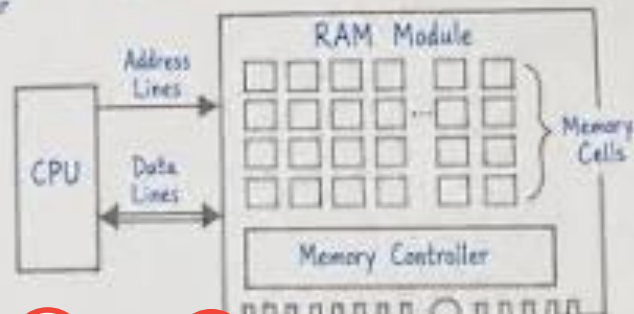
RAM Overview

RAM is the main memory of a computer used to store data and instructions temporarily while the CPU is processing tasks.

Key features:

- Volatile memory
- High-speed access
- Directly connected to CPU

RAM Architecture

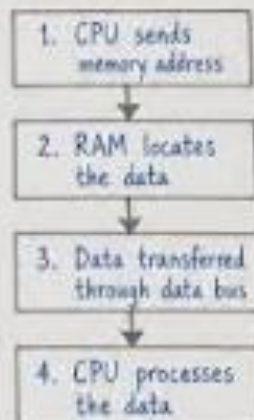


Types of RAM

- SRAM (Static RAM) — very fast memory used in cache.
- DRAM (Dynamic RAM) — main system memory used in computers.

Feature	SRAM	DRAM
Full Form	Static RAM	Dynamic RAM
Speed	Very Fast	Relatively Slower
Cost	High	Low
Density	Lower	Higher
Power Use	Higher	Lower
Use	Cache Memory	Main Memory

Memory Operation



DDR Memory Evolution

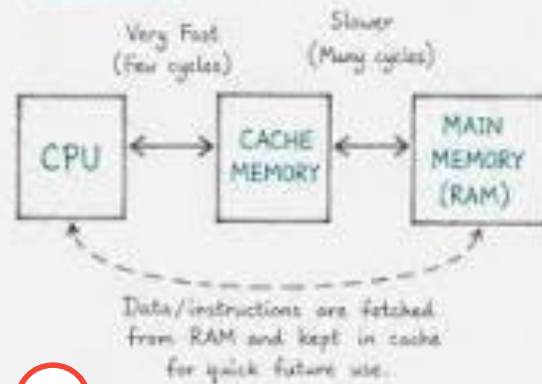


Cache Memory Basics

Cache memory is a small, very fast memory located between the CPU and the main memory (RAM). It stores copies of the data and instructions that the CPU uses most often.

Why is Cache Needed?

- The CPU is extremely fast.
- Main memory (RAM) is much slower.
- Without cache, the CPU would have to wait a lot for data, wasting time.
- Cache reduces this waiting time by keeping frequently used data close to the CPU.



How Does Cache Help the CPU?

- When the CPU needs data or an instruction, it first checks the cache.
- If it's found in the cache → it's a "Hit" → CPU gets it instantly.
- If it's not found → a "Miss" → data is brought from RAM, stored in cache, and then given to the CPU.
- Since recently/frequently used items stay in cache, most requests are served quickly (high hit rate), improving performance.

Levels of Cache

- L1 Cache: Smallest but fastest. Located inside the CPU core. Used first.
- L2 Cache: Bigger but slightly slower than L1. Also on or near the CPU.
- L3 Cache: Even bigger, shared by all cores (in many CPUs). Slower than L1/L2 but faster than RAM.

Quick Comparison

Level	Size	Speed	Location	Typical Use
L1 Cache	Smallest	Fastest	Inside CPU core	Used first
L2 Cache	Medium	Slower than L1	On/near CPU	Backup to L1
L3 Cache	Largest	Slower than L2	Shared (on chip)	Last cache before RAM
Main Memory (RAM)	Very large	Slowest	On motherboard	Stores all data and programs

REMEMBER

Cache keeps the data and instructions the CPU uses most often.

More cache (and faster cache) = less waiting for RAM = better performance!